

Sequences: Convergence and Divergence

Theorem (Sandwich theorem)

Let $\{u_n\}, \{v_n\}, \{w_n\}$ be three sequences of real numbers and there is a natural number m such that

$$u_n < v_n < w_n \quad \text{for all } n \geq m.$$

If $\lim u_n = \lim w_n = l$ then $\{v_n\}$ is convergent and $\lim v_n = l$.

Proof of Theorem

Let $\epsilon > 0$. From convergence of $\{u_n\}$ and $\{w_n\}$, there exist k_1, k_2 such that

$$|u_n - l| < \epsilon \quad (n \geq k_1), \quad |w_n - l| < \epsilon \quad (n \geq k_2).$$

Let $k_3 = \max\{k_1, k_2\}$. Then

$$l - \epsilon < u_n < v_n < w_n < l + \epsilon \quad (n \geq \max\{k_3, m\}).$$

Hence $|v_n - l| < \epsilon$, proving $\lim v_n = l$. □

If $u_n \leq v_n \leq w_n$ for all $n \geq m$ and $\lim u_n = \lim w_n = l$, then $\lim v_n = l$.

Worked Example 1

Prove that

$$\lim_{n \rightarrow \infty} \frac{3n^2 + 2n + 1}{n^2 + 1} = 3.$$

We write $u_n = 3 + \frac{2}{n} + \frac{1}{n^2}$, $v_n = 1 + \frac{1}{n^2}$.

Then $\lim u_n = 3$, $\lim v_n = 1$. Hence $\lim \frac{u_n}{v_n} = \frac{3}{1} = 3$.

Worked Example 2

Prove that

$$\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0.$$

$$\sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}}.$$

Let $u_n = \frac{1}{\sqrt{n}}$, $v_n = \frac{\sqrt{n}}{\sqrt{n+1} + \sqrt{n}}$. Then $u_n \rightarrow 0$, $v_n \rightarrow \frac{1}{2}$. Hence limit is 0.

Worked Example 3

Prove that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n+1}} + \cdots + \frac{1}{\sqrt{n+n}} \right) = 1.$$

$$\frac{n}{\sqrt{n^2 + n}} < u_n < \frac{n}{\sqrt{n^2 + 1}}.$$

Thus

$$\frac{1}{\sqrt{1 + \frac{1}{n}}} < u_n < \frac{1}{\sqrt{1 + \frac{1}{n^2}}}.$$

Both bounds $\rightarrow 1$, hence $\lim u_n = 1$ by Sandwich theorem.

5.6 Null sequence

A sequence $\{u_n\}$ is called a *null sequence* if $\lim u_n = 0$.

5.7 Divergent sequence

A sequence $\{f(n)\}$ is said to *diverge to* ∞ if for any $G > 0$ there is k such that

$$f(n) > G \quad (n \geq k).$$

We write $\lim f(n) = \infty$.

Similarly, $\{f(n)\}$ diverges to $-\infty$ if for any $G > 0$ there is k such that

$$f(n) < -G \quad (n \geq k).$$

Divergent Sequences

In this case we write $\lim f(n) = -\infty$ and also say that the sequence $\{f(n)\}$ tends to $-\infty$.

A real sequence $\{f(n)\}$ is said to be a **properly divergent sequence** if it either diverges to ∞ , or diverges to $-\infty$.

Theorem

A sequence diverging to ∞ is unbounded above but bounded below

Proof. Let a sequence $\{f(n)\}$ diverge to ∞ . Then for each pre-assigned positive number G there exists a natural number k such that $f(k) > G$. Therefore there does not exist a real number B such that $f(n) \leq B$ holds for all $n \in \mathbb{N}$. In other words, $\{f(n)\}$ is unbounded above. Let $G > 0$. Then there exists a natural number k such that $f(n) > G$ for all $n \geq k$. Let $b = \min\{f(1), f(2), \dots, f(k-1), G\}$. Then $f(n) \geq b$ for all $n \in \mathbb{N}$. This proves that the sequence $\{f(n)\}$ is bounded below.

Note. A sequence unbounded above but bounded below may not diverge to ∞ . For example, let us consider the sequence $\{f(n)\}$ where

$$f(n) = \begin{cases} n & \text{if } n \text{ is odd} \\ 1 & \text{if } n \text{ is even} \end{cases}$$

The sequence is unbounded above and bounded below, 0 being a lower bound.

The sequence does not diverge to ∞ , because for a pre-assigned positive number G there does not exist a natural number k such that $f(n) > G$ holds for all $n \geq k$.

Theorem

Theorem

A sequence diverging to $-\infty$ is unbounded below but bounded above.

Note. A sequence unbounded below but bounded above may not diverge to $-\infty$. For example, let us consider the sequence $\{f(n)\}$ where

$$f(n) = \begin{cases} -n & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

The sequence is unbounded below and bounded above, 1 being a upper bound. But this sequence does not diverge to $-\infty$

Examples.

- 1 The sequence $\{2^n\}$ diverges to ∞ .
- 2 The sequence $\{-n^2\}$ diverges to $-\infty$.
- 3 The sequence $\{(-1)^n\}$ is a bounded sequence, but not convergent. It is an oscillatory sequence of finite oscillation.
- 4 The sequence $\{(-1)^n n\}$ is an unbounded sequence, and it is not properly divergent. It is an oscillatory sequence of infinite oscillation.

5.8. Some important limits.

$\lim r^n = 0$ if $|r| < 1$.

Case 1. $r = 0$. In this case the sequence is $\{0, 0, 0, \dots\}$. The sequence converges to 0.

That is, $\lim r^n = 0$ when $r = 0$.

Some Important Limits (contd.)

Case 2. $r \neq 0$ and $|r| < 1$.

$\frac{1}{|r|} > 1$, since $|r| < 1$. Let $\frac{1}{|r|} = a + 1$ where $a > 0$.

$$|r^n - 0| = |r|^n = r^n = \frac{1}{\left(\frac{1}{r}\right)^n} = \frac{1}{(a+1)^n}.$$

We have $(1+a)^n > na$ for all $n \in \mathbb{N}$. So $\frac{1}{(a+1)^n} < \frac{1}{na}$ for all $n \in \mathbb{N}$.

Let $\epsilon > 0$. Then $|r^n - 0| < \epsilon$ holds if $n > \frac{1}{a\epsilon}$. Let $k = \left\lceil \frac{1}{a\epsilon} \right\rceil + 1$. Then k is a natural number and $|r^n - 0| < \epsilon$ for all $n \geq k$.

Since ϵ is arbitrary, $\lim r^n = 0$.

Combining the cases, $\lim r^n = 0$ if $|r| < 1$.

Limit of $a^{1/n}$

$$\lim a^{1/n} = 1 \text{ if } a > 0.$$

Corollary 1.

If $\lim x_n = l$ and $a > 0$, then $\lim a^{x_n} = a^l$.

$$\text{Let } x_n = 3 + \frac{1}{n}, \quad a = 2.$$

Then clearly

$$\lim_{n \rightarrow \infty} x_n = 3.$$

Now consider

$$2^{x_n} = 2^{3 + \frac{1}{n}} = 2^3 \cdot 2^{1/n} = 8 \cdot 2^{1/n}.$$

Since $\lim_{n \rightarrow \infty} 2^{1/n} = 1$, we get

$$\lim_{n \rightarrow \infty} 2^{x_n} = 8 \cdot 1 = 8 = 2^3.$$

Corollary 2.

If $\lim x_n = l$, then $\lim e^{x_n} = e^l$.

$$\text{Let } x_n = 2 + \frac{1}{n}.$$

Then

$$\lim_{n \rightarrow \infty} x_n = 2.$$

Now consider

$$e^{x_n} = e^{2 + \frac{1}{n}} = e^2 \cdot e^{1/n}.$$

Since $\lim_{n \rightarrow \infty} e^{1/n} = 1$, we obtain

$$\lim_{n \rightarrow \infty} e^{x_n} = e^2 \cdot 1 = e^2.$$

Theorem

If $\lim x_n = 0$, then $\lim \log(1 + x_n) = 0$.

$$\text{Let } x_n = \frac{1}{n}.$$

Then

$$\lim_{n \rightarrow \infty} x_n = 0.$$

$$\lim_{n \rightarrow \infty} \log\left(1 + \frac{1}{n}\right) = 0.$$

Important Limit

Theorem.

If $u_n > 0$ and $\lim u_n = u > 0$ for all $n \in \mathbb{N}$ and $\lim v_n = v$, then $\lim(u_n)^{v_n} = u^v$.

$$\text{Let } u_n = 2 + \frac{1}{n}, \quad v_n = 3 + \frac{(-1)^n}{n}.$$

Clearly,

$$\lim_{n \rightarrow \infty} u_n = 2 > 0, \quad \lim_{n \rightarrow \infty} v_n = 3.$$

Now consider

$$(u_n)^{v_n} = \left(2 + \frac{1}{n}\right)^{3 + \frac{(-1)^n}{n}}.$$

As $n \rightarrow \infty$,

$$(u_n)^{v_n} \longrightarrow 2^3 = 8.$$

Hence,

$$\lim (u_n)^{v_n} = 2^3 = 8.$$

Limit Example

$$\lim n^{1/n} = 1.$$

Behaviour of Sequence r^n

For different real values of r :

Case 1. $r > 1$. In this case the sequence diverges to ∞ .

Case 2. $r = 1$. The sequence is $\{1, 1, 1, \dots\}$ and converges to 1.

Case 3. $|r| < 1$. The sequence converges to 0.

Case 4. $r = -1$. The sequence is $\{-1, 1, -1, 1, \dots\}$. It is bounded but not convergent (oscillatory).

Case 5. $r < -1$. Let $r = -s$, then $s > 1$. The sequence is $\{(-1)^n s^n\}$. It is unbounded, neither diverges to ∞ nor $-\infty$, but is oscillatory.

Theorem

Theorem

Let $\{u_n\}$ be a sequence of positive real numbers such that $\lim \frac{u_{n+1}}{u_n} = l$.

(i) If $0 \leq l < 1$ then $\lim u_n = 0$.

$$\text{Let } u_n = \left(\frac{1}{2}\right)^n.$$

Then

$$\frac{u_{n+1}}{u_n} = \frac{\left(\frac{1}{2}\right)^{n+1}}{\left(\frac{1}{2}\right)^n} = \frac{1}{2}.$$

Hence

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \frac{1}{2} = \ell < 1.$$

Therefore, by the result,

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^n = 0.$$

5.9 Monotone Sequence.

A real sequence $\{f(n)\}$ is said to be a *monotone increasing sequence* if

$$f(n+1) \geq f(n) \quad \text{for all } n \in \mathbb{N}.$$

A real sequence $\{f(n)\}$ is said to be a *monotone decreasing sequence* if

$$f(n+1) \leq f(n) \quad \text{for all } n \in \mathbb{N}.$$

A real sequence $\{f(n)\}$ is said to be a *monotone sequence* if it is either a monotone increasing or monotone decreasing sequence.

Monotone Sequence (Strict and Ultimate)

Note.

If $f(n+1) > f(n)$ for all $n \in \mathbb{N}$, then $\{f(n)\}$ is a **strictly monotone increasing sequence**.

If $f(n+1) < f(n)$ for all $n \in \mathbb{N}$, then $\{f(n)\}$ is a **strictly monotone decreasing sequence**.

If for some natural number m , $f(n+1) \geq f(n)$ for all $n \geq m$, then $\{f(n)\}$ is said to be an **ultimately monotone increasing sequence**.

If for some natural number m , $f(n+1) \leq f(n)$ for all $n \geq m$, then $\{f(n)\}$ is said to be an **ultimately monotone decreasing sequence**.

Examples of Monotone Sequences

Examples.

1. Let $f(n) = 2^n$, $n \geq 1$. Then $f(n+1) > f(n)$ for all $n \in \mathbb{N}$.

$\Rightarrow \{f(n)\}$ is monotone increasing and strictly monotone.

2. Let $f(n) = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}$, $n \geq 1$.

Then

$$f(n+1) - f(n) = \frac{1}{2n+1} + \frac{1}{2n+2} + \cdots + \frac{1}{n+1} - \frac{1}{n+1}.$$

$\Rightarrow \{f(n)\}$ is monotone increasing and strictly monotone.

Examples (continued)

3. Let $f(n) = \frac{1}{n}$, $n \geq 1$.

Then $f(n+1) - f(n) < 0$ for all $n \in \mathbb{N}$.

$\Rightarrow \{f(n)\}$ is monotone decreasing and strictly monotone.

4. The sequence $\{(-2)^n\}$ is neither monotone increasing nor monotone decreasing. Therefore it is not a monotone sequence.

Theorem

A monotone increasing sequence, if bounded above, is convergent and it converges to the least upper bound.

Proof. Let $\{f(n)\}$ be monotone increasing and bounded above. Let M be its least upper bound.

Then: (i) $f(n) \leq M$ for all $n \in \mathbb{N}$

(ii) For a pre-assigned $\epsilon > 0$, there exists a natural number k such that $f(k) > M - \epsilon$.

Since $\{f(n)\}$ is increasing:

$$M - \epsilon < f(k) \leq f(k+1) \leq f(k+2) \leq \dots \leq M.$$

That is, $M - \epsilon < f(n) < M + \epsilon$ for all $n \geq k$.

Thus $\{f(n)\}$ is convergent and $\lim f(n) = M$.

Theorem

A monotone decreasing sequence, if bounded below, is convergent and it converges to the greatest lower bound.

(Proof similar to the previous theorem.)

Theorem

A monotone increasing sequence that is unbounded above diverges to ∞ .

Proof. Let $\{f(n)\}$ be monotone increasing and unbounded above. Then, for any $G > 0$, there exists $k \in \mathbb{N}$ such that $f(k) > G$. Since $\{f(n)\}$ is increasing,

$$G < f(k) \leq f(k+1) \leq f(k+2) \leq \dots$$

Thus $\lim f(n) = \infty$.